

WASP-10b

R-0817

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_171130 · created 2026-07-28T13:38:05+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458087.6838 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.205 +/- 0.059
Transit depth (Rp/Rs)^2	4.19 +/- 2.4 [%]
Orbital Inclination (inc)	85.61 +/- 1.85
Airmass coefficient 1 (a1)	1.21 +/- 0.11
Airmass coefficient 2 (a2)	-0.163 +/- 0.088
Scatter in the residuals of the lightcurve fit is	9.11 %
Best Comparison Star	#3 - [262, 348]
Optimal Aperture	2.16
Optimal Annulus	9.55
Transit Duration (day)	0.058 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.205 ± 0.059 vs 0.1592 (deviation 0.78σ)
Duration fit vs published	0.058 d vs 0.0946 d (deviation 1.08σ)
Mid-time O–C	-6.1 min
Depth SNR	3.5
Residual scatter	9.11 %

Light curve

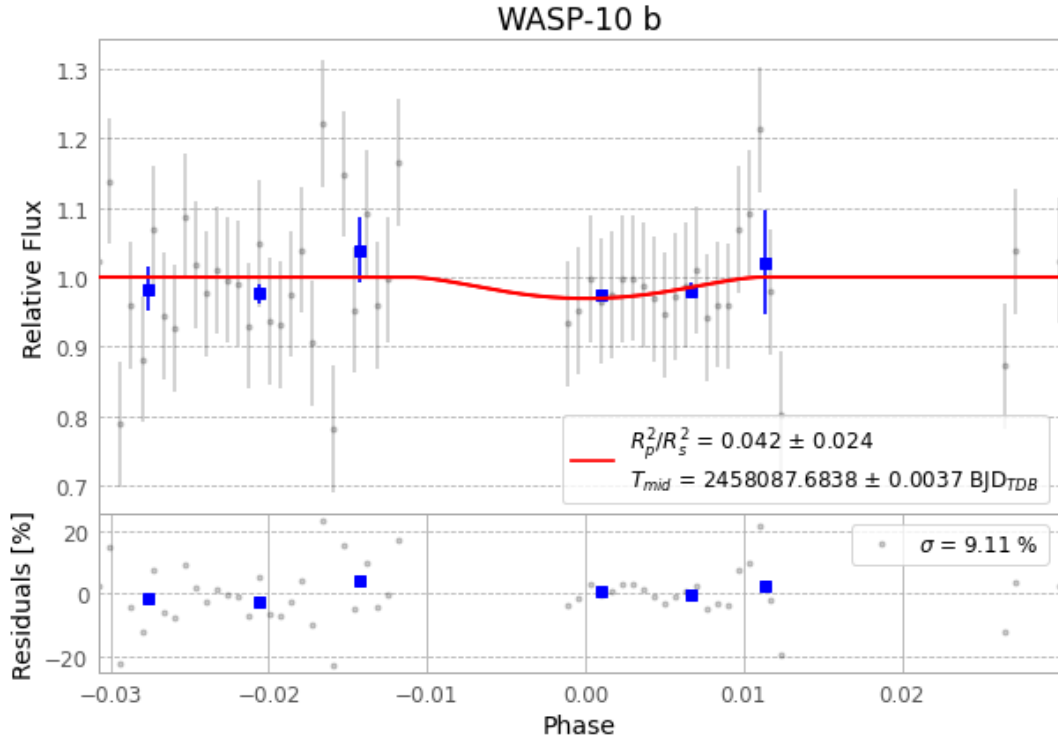


Figure 1 — FinalLightCurve_WASP-10 b_2017-11-29.png (SHA-256 8788f07c695b3bbc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c449200d08ea746e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

97 FITS frames (64 MB), WASP-10171130020312.FITS → WASP-10171130065112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-171130.log (31011dfe482c...), FinalLightCurve_WASP-10 b_2017-11-29.png (8788f07c695b...), FinalLightCurve_WASP-10 b_2017-11-29.csv (5f7062dfd695...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 13:38Z.